

LFER Part 2: Substituents with More Punch. Plots of Data from Part 1

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This document was produced using the $\text{\LaTeX}$ typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*. Diagrams were edited in *Affinity Publisher*

The handout from the previous class meeting features several tables of data. We used the classic Hammett σ parameters in the analysis but today we will replot some of these data sets using the Brown-Okamoto σ^+ parameters when appropriate. You can use the Jupyter notebooks accessible via the course web book to make your own versions of these plots.

Table 4: Benzyl Chloride with Water and with Hydroxide Ion

Should we use the Brown-Okamoto σ^+ parameters for the S_N2 substitution mechanism in solvolysis of benzyl chloride? The data does show that the p-OMe point in the acid hydrolysis conditions is well above the line fit. However, when one considers the scatter in the data in general, it is not an outlier. If the transition state did have a highly electron-deficient carbon attached to the benzene ring that could accept electron density from a para substituent then it would be acceptable to use σ^+ rather than σ . Generally S_N2 mechanisms do not benefit from direct resonance stabilization at the reaction centre because the electron density is not reduced enough at the transition state, even when a slight positive charge is developing. So we will conclude that we do not need to reconsider the plot for table 4.

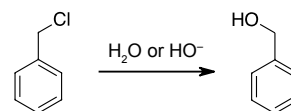


Figure 1: Hydrolysis of benzyl chloride

Table 6: Solvolysis of Diphenylmethylene Chloride

This solvolysis of diphenylmethylene chloride does proceed via an S_N1 mechanism because the two phenyl rings electronically stabilize the cationic intermediate and sterically destabilize the transition state needed for S_N2 . Since a very electron deficient carbon is forming in the transition state there will be the opportunity to share the charge via direct resonance contributions from para substituents that are resonance donors. We should have used the σ^+ parameters. I chose to plot the data for solvolysis in ethanol. You can plot other data sets from table 6 by using the [Python notebook available via Colab](#).

We see a large improvement in the correlation of $\log K_{obs}$ vs. σ^+ . The slope did not change greatly but we clearly see that parameters that include resonance donation through to the reaction centre are better in this case.

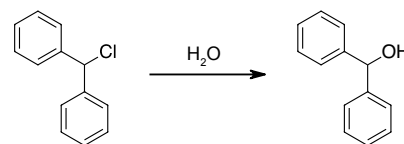


Figure 2: Solvolysis of Diphenylmethylene Chloride

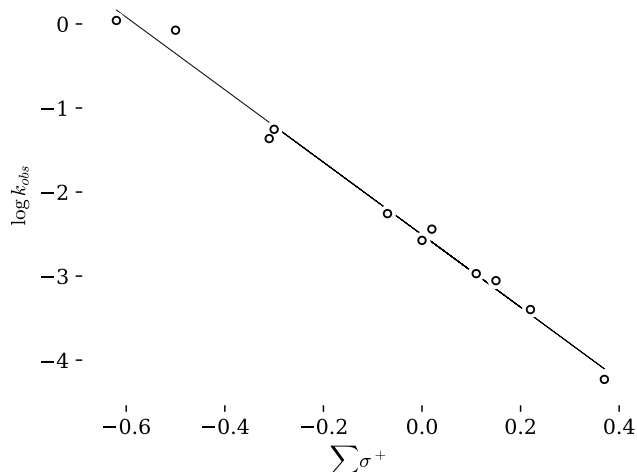


Figure 3: Hammett plot for solvolysis of diphenylmethylene chloride. Correlation is poor but we still see the trend. Perhaps we should use different σ parameters?

$$\rho = -4.31 \pm 0.15, R^2 = 0.99$$

Table 7: Solvolysis of Trityl Chloride

In table 7 of the previous class handout we presented data for the rate of substituted trityl chlorides reacting with water in acetonitrile at 25 °C. We used Hammett σ parameters in the previous class but now we know that it would be more appropriate to use the Brown-Okamoto σ^+ parameters. As you can see in the new plot, the correlation improved slightly but is not obviously better. Quality of fit is not the reason to choose σ^+ rather than σ values, we choose based on our hypothesis of the mechanism.

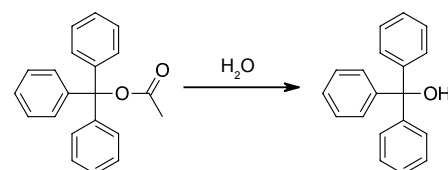


Figure 4: Solvolysis of Trityl Chloride

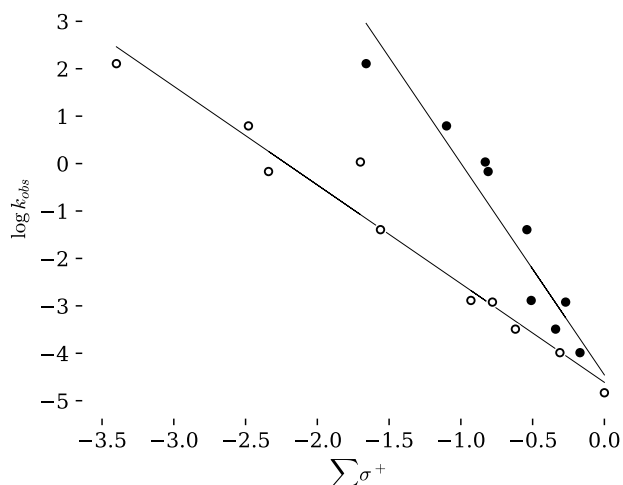


Figure 5: Hammett plot for solvolysis of trityl chloride. Both correlations with Hammett σ and Brown-Okamoto σ^+ values are shown for comparison. Both look OK but note the difference in slope.

Hammett σ parameters,
closed circles
 $\rho = -4.47 \pm 0.44, R^2 = 0.93$

Brown-Okamoto σ^+ parameters,
open circles
 $\rho = -2.08 \pm 0.14, R^2 = 0.96$

I chose to plot the data for solvolysis in 90% acetonitrile. You can plot other data sets from table 7 by using the [Python notebook available via Colab](#).

Analysis & Comments

Let us consider the two cases that were plotted above. In the solvolysis of diphenylmethyle chloride we observe a better correlation with σ^+ , as expected. In the solvolysis of triphenylmethyle chloride (trityl chloride) we observed marginal improvement in correlation but a large difference in the observed value of the reaction constant, ρ , when we used σ^+ parameters.

Conclusions

Both mechanisms proceed via the S_N1 mechanism. We observed that the trityl chloride was less affected by the electronic influence of the substituent (as scored by σ^+) compared to the diphenylmethyle chloride. One explanation for this is that the trityl chloride phenyl groups are not able to all occupy the same plane. They must twist to fit around the planar sp^2 carbocation centre. This twist reduces the overlap of the π -systems of the phenyl rings and also the resonance connection with substituents. The effect of resonance is significant, but is reduced accordingly. Also, if the charge is shared among three phenyl rings then there is a less intense positive charge at the reaction centre and ρ would be correspondingly smaller in magnitude.

I performed a Yukawa-Tsuno analysis and found then a r value of around 0.7 resulted in the best fit (although, as you can see, a range of $0.4 > r > 1.0$ could easily be accepted.) So this twist in the phenyl groups reduces the resonance effectiveness by about 30%. Do you believe it? the code for this analysis is [is available via Colab](#).

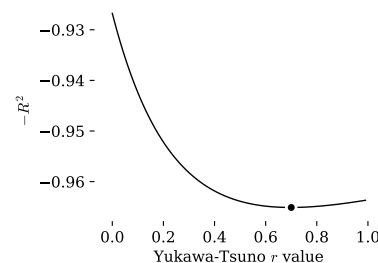


Figure 6: Optimization of Yukawa-Tsuno r -value for solvolysis of trityl chloride. the best correlation was found with $r = 0.70$

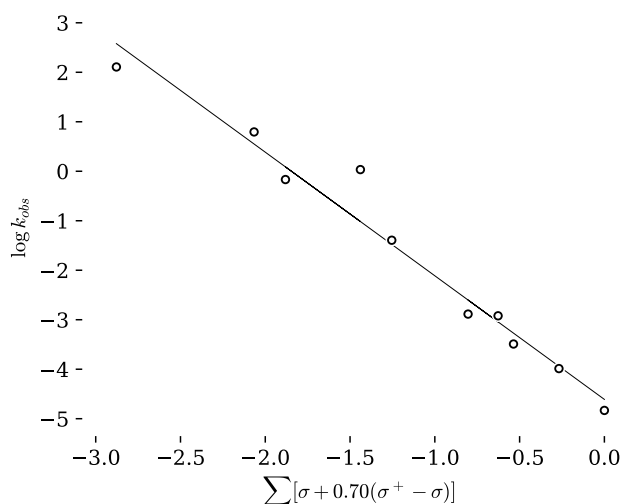


Figure 7: Hammett plot for solvolysis of trityl chloride using the Yukawa-Tsuno method with $r = 0.70$.

$$\rho = -2.50 \pm 0.17, R^2 = 0.97$$